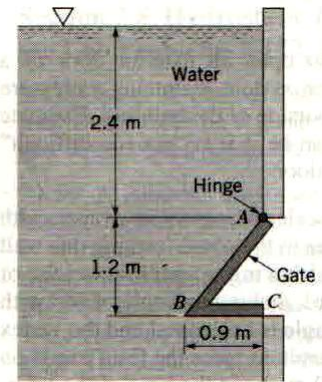
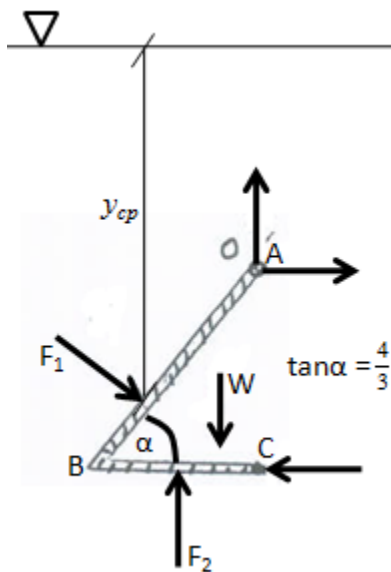


$$\text{Use } g = 9.81 \text{ m/s}^2 \text{ and } \rho_{\text{water}} = 1000 \text{ kg/m}^3$$

- Q2. A gate having the cross section shown in FigQ2 closes an opening 1.5 m wide and 1.2 m high in a water reservoir. The gate weighs 2.2 kN and its center of gravity is 0.3 m to the left of AC and 0.6 m above BC. Determine the horizontal reaction that is developed on the gate at C.



### FBD of Gate



$$F_1 = \rho_{\text{water}} g \bar{y} A = 9810 \left( 2.4 + \frac{1.2}{2} \right) (1.5 \times 1.5) = 66.22 \text{ kN}$$

$$\text{To locate } F_1 \Rightarrow y_{cp} = \bar{y} + \frac{I_G \sin^2 \alpha}{A \bar{y}} \Rightarrow y_{cp} = 3 + \frac{\frac{1}{12} (1.5 \times 1.5^3) (0.8)^2}{(1.5 \times 1.5) 3} = 3.04 \text{ m}$$

$$F_2 = P_2 A_2 = \rho g \bar{h} A_{BC} = 9810 (2.4 + 1.2) (1.5 \times 0.9) = 47.68 \text{ kN}$$

$$\sum M_A = 0 \Rightarrow F_1 \left( \frac{y_{cp} - 2.4}{\sin \alpha} \right) + W(0.3) - F_2(0.45) - R_c(1.2) = 0$$

$$R_c = 26.82 \text{ kN}$$